

ORIGINAL CONTRIBUTION

Body Weight in Relation to the Effectiveness of Intravenous Alteplase in Acute Ischemic Stroke: The ENCHANTED Trial

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BACKGROUND: The prognostic significance of body weight in patients receiving thrombolysis for acute ischemic stroke remains uncertain. This study aimed to determine whether body weight modifies the comparative effects of low- versus standard-dose intravenous alteplase and to evaluate the associations of body weight and clinical outcomes in patients with acute ischemic stroke.

METHODS: A post hoc analysis of the ENCHANTED trial (Enhanced Control of Hypertension and Thrombolysis Stroke Study), an international partial-factorial randomized controlled trial comparing low-dose (0.6 mg/kg) versus standard-dose (0.9 mg/kg) intravenous alteplase and intensive (target systolic blood pressure 130–140 mmHg) versus guideline-recommended (target systolic blood pressure <180 mmHg) treatment. The primary outcome was poor functional outcome at 90 days, defined as a modified Rankin Scale score of 2 to 6 (versus 0–1). For the first objective, the exposure was defined as alteplase dose and body weight, whereas for the second objective, body weight alone was considered as the exposure. Logistic regression models were performed to analyze the association between body weight and outcome. The association shape between treatment effect and body weight was evaluated using restricted cubic splines.

RESULTS: Among 3206 patients (mean age, 66.6 years; female 37.9%) in the alteplase-dose arm, body weight did not significantly modify the comparative effects of low- versus standard-dose alteplase on functional recovery ($P_{\text{interaction}}=0.60$), death ($P_{\text{interaction}}=0.56$), or symptomatic intracerebral hemorrhage ($P_{\text{interaction}}=0.46$). In the full cohort of 4458 patients (mean age 66.7 years; female 37.8%), a nonlinear relationship was observed between continuous body weight and 90-day modified Rankin Scale score, with an inflection point at ≈ 74 kg. Patients weighing <74 kg had a significantly higher likelihood of poor functional outcome (odds ratio, 1.41 [95% CI, 1.06–1.87]; $P=0.017$) and death (1.65 [95% CI, 1.04–2.61]; $P=0.03$), but not an increase in symptomatic intracerebral hemorrhage.

CONCLUSIONS: There was no evidence of effect modification by body weight on the effectiveness of low- versus standard-dose intravenous alteplase in patients with acute ischemic stroke. Low body weight predicts a poor outcome but not an increased risk of symptomatic intracerebral hemorrhage after thrombolysis treatment for acute ischemic stroke.

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Key Words: blood pressure ■ body weight ■ cerebral hemorrhage ■ prognosis ■ stroke

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Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
BP	blood pressure
ENCHANTED	Enhanced Control of Hypertension and Thrombolysis Stroke Study
mRS	modified Rankin Scale
rt-PA	recombinant tissue-type plasminogen activator
sICH	symptomatic intracerebral hemorrhage

Intravenous thrombolysis with rt-PA (recombinant tissue-type plasminogen activator, or alteplase) has been the cornerstone of treatment of acute ischemic stroke (AIS) for 3 decades.¹ Although its efficacy is well established, treatment response may vary according to individual patient factors such as body weight.² Body weight influences both baseline prognosis and clinical outcomes, with overweight patients often demonstrating better recovery, a phenomenon that has been termed the obesity paradox.^{3–8} Conversely, low body weight, often associated with frailty and comorbidities, tends to have less favorable outcomes,^{7,9} although the supporting evidence is mixed and controversial.¹⁰

Although a fixed dose of 0.9 mg/kg (maximum 90 mg) intravenous alteplase is approved for the treatment of AIS,¹¹ body weight appears to influence its efficacy on functional recovery but not for the odds of symptomatic intracerebral hemorrhage (sICH).^{12–14} To determine whether a lower dose of intravenous alteplase might improve the safety and efficacy of reperfusion treatment, the alteplase-dose arm ENCHANTED trial (Enhanced Control of Hypertension and Thrombolysis Stroke Study) was undertaken to compare low-dose (0.6 mg/kg) with standard-dose (0.9 mg/kg) alteplase in AIS.^{15–17} This international clinical trial in predominantly Asian patients found that low-dose alteplase was not shown to meet the predefined noninferiority margin compared with standard-dose alteplase with respect to the conventional binary end point of death or disability, defined by scores of 2 to 6 on the modified Rankin Scale (mRS) at 90 days.¹⁷ However, the study did show that low-dose alteplase was noninferior for overall functional recovery through an ordinal analysis of the mRS score and resulted in significantly less sICH than standard-dose alteplase.^{15–17} It is plausible that low-dose alteplase may be preferable in certain types of AIS patients. We hypothesized that low-dose alteplase may be a safer treatment in patients at high-risk of sICH,¹⁸ such as those with low body weight. We aimed first to investigate whether body weight modified the effect of low-dose versus standard-dose alteplase. Next, we quantified the associations of body

weight and clinical outcomes among patients with AIS enrolled in the ENCHANTED trial.

METHODS

Data Sharing Statement

Data of individually deidentified participants used in this study may be permitted to be shared by formal request with a protocol and statistical plan from any qualified investigator to the Research Office of The George Institute for Global Health, Australia. A tailored data set specific to the research question can be shared for 6 months, and the data can be accessed only by qualified statisticians for the proposed analysis.

Participants

ENCHANTED was an international, multicenter, prospective, randomized, open-label, blinded-end point trial, the details of which are reported elsewhere.^{15–17} In brief, 3310 patients with a diagnosis of AIS confirmed on brain imaging and fulfilling standard criteria for thrombolysis treatment, including having a symptom onset within 4.5 hours, were randomly assigned to receive low-dose (0.6 mg/kg; 15% as bolus, 85% as infusion over 1 hour) or standard-dose (0.9 mg/kg; 10% as bolus, 90% as infusion over 1 hour) alteplase. Body weight was determined by clinicians according to usual clinical practice, whether directly measured or estimated based on a report. The blood pressure (BP) arm,¹⁹ conducted between March 3, 2012 and April 30, 2018, included a total 2227 patients with elevated systolic BP (SBP; 150–180 mmHg) where the attending investigator was uncertain of the benefits and risks of intensity of BP control, immediately after thrombolytic treatment and for the next 72 hours (or hospital discharge or death if this occurred earlier). Although there was no specified upper SBP level, international guidelines recommend patients have a SBP ≤ 185 mmHg before administration of intravenous alteplase.²⁰ Participants were randomly assigned to a strategy of intensive BP lowering (target SBP 130 to 140 mmHg within 60 minutes of randomization) or guideline-recommended BP lowering (target SBP < 180 mmHg) after the commencement of intravenous alteplase.

The study protocol was approved by the appropriate ethics committee at each participating center, and written informed consent was obtained from each patient or an appropriate surrogate.

Procedures

In this post hoc exploratory secondary analysis of the ENCHANTED trial, key demographic and clinical characteristics were recorded at the time of enrollment. Stroke severity was measured using the National Institutes of Health Stroke Scale score at baseline, 24 hours, and at day 7 (or earlier on discharge from hospital). Uncompressed digital images of all baseline and follow-up computed tomography, magnetic resonance imaging, and angiogram images were collected in DICOM format on a CD-ROM identified only with the patient's unique study number and uploaded by a special purpose-built Web-based system for central analysis at The George Institute for Global Health. All brain scans with an intracranial hemorrhage were reviewed by at least 2 independent assessors,

blind to clinical data, treatment, and date and sequence of scan, using MISTar version 3.2 (Apollo Medical Imaging Technology, Melbourne, Victoria, Australia). Assessors graded any hemorrhage as intracerebral, subarachnoid, intraventricular, subdural, or other; sICH being graded across all standard definitions (SITS-MOST, NINDS, ECASS2, ECASS3 and IST3). This article follows the STROBE reporting guideline, with all checklist items considered in the reporting of study design, data analysis, and results to enhance transparency and reproducibility.²¹

Outcomes

The primary outcome was functional status at 90 days. The proportional odds assumption for ordinal mRS score analysis was tested and not fulfilled;^{22,23} the outcome was analyzed as a dichotomous variable defined by mRS scores of 2 to 6 versus 0 to 1.²⁴ Secondary outcomes were any intracerebral hemorrhage reported by investigators with or without central adjudication of relevant brain imaging by the criteria of NINDS within 7 days after randomization, mRS scores 3 to 6, and death within 90 days.

Statistical Analysis

The treatment effects of low-dose versus standard-dose alteplase on mRS score, death, and sICH were estimated using logistic regression models. Heterogeneity of treatment effect across body weight was assessed by including treatment-by-weight interaction terms, with body weight modelled as a continuous variable (1 degree of freedom). We further assessed the shape of associations using restricted cubic splines. The models were stratified by sex for sensitivity analysis. Data are reported as odds ratios and 95% CIs. The associations of body weight (modelled as both a continuous variable and as categories of ≤ 50 , 51–60, 61–70, 71–80, and >80 kg, chosen to ensure adequate group sizes) with death or disability, death, and sICH were assessed using logistic regression models adjusted for baseline covariates.

Models for both analyses were adjusted for baseline covariates selected a priori based on clinical relevance and prior literature. In addition, variables exhibiting notable baseline imbalance between groups were included to further control for potential confounding. These covariates including randomized treatment allocation; time from onset to randomization; age; sex; ethnicity; baseline SBP; baseline National Institutes of Health Stroke Scale score; history of stroke, hypertension, atrial fibrillation, diabetes, valvular heart disease, and hypercholesterolemia; current treated hypertension; current smoking status; premorbid use of anticoagulants, other antiplatelet agents, and statins or other lipid-lowering therapies; and presumed stroke cause. Two-sided *P* values are reported, with $P < 0.05$ considered statistically significant. All analyses were undertaken using R version 3.4.2 (R Foundation for Statistical Computing) with the “rms” and “marginaleffects” packages.^{25,26}

RESULTS

Effects of Low- Versus Standard-Dose Alteplase by Body Weight

From March 2012 to August 2015, a total of 3310 patients were randomized (Figure S1), of whom 1654

patients were assigned to low-dose alteplase and 1643 to standard-dose alteplase.¹⁷ Of these patients, 3206 (mean age, 66.6 [SD 12.8] years; female, 37.9%) had available information on body weight for analyses. Table 1 shows the characteristics of patients by body weight; patients with low body weight were significantly more likely to be female, Asian, had greater neurological impairment (higher National Institutes of Health Stroke Scale scores), and less likely to have the comorbidities of hypertension, valvular or other heart disease, diabetes, and hypercholesterolemia, and prior use of aspirin or statin therapy. In addition, patients with low body weight were more likely to have had atrial fibrillation and cardioembolic stroke.

The estimated treatment effect on 90-day poor functional outcome (mRS score, 2–6) showed nominal variation across the body weight distribution (Figure 1). The direction of effect was broadly consistent, whereas the magnitude of the estimated effect differed across weight values, with wider CIs at the extremes. There were no differences in the comparison of effects of low-dose versus standard-dose alteplase according to body weight for the binary mRS score outcome (Wald $\chi^2=0.68$; $P_{\text{interaction}}=0.60$), any intracerebral hemorrhage (Wald $\chi^2=0.58$; $P_{\text{interaction}}=0.56$), or mortality (Wald $\chi^2=0.22$; $P_{\text{interaction}}=0.49$).

Prognostic Significance of Body Weight

There were 4458 patients who participated in both arms of ENCHANTED (mean age, 66.7 [12.7] years; female, 37.8%; Asian ethnicity, 65.9%) and had data available for analysis (Figure 2; Table S1). After adjusting for key baseline covariates, there was an asymmetrical quasi-U-shaped association between continuous body weight and binary mRS score at 90 days, with an inflection point at ≈ 75 kg (Figure 3). This pattern was consistent when body weight was analyzed categorically. Compared with the reference group (71–80 kg), patients weighing <50 kg had a significantly higher likelihood of having a poor functional outcome (adjusted odds ratio, 1.41 [95% CI, 1.06–1.87]). However, there were no significant differences across the other weight categories (1.04 [95% CI, 0.84–1.27] for 51–60 kg, 1.13 [95% CI, 0.94–1.36] for 61–70 kg, and 1.06 [95% CI, 0.86–1.32] for >80 kg; Table 2).

After adjustment for key baseline covariates, an asymmetrical quasi-U-shaped association was evident between continuous body weight and mortality at 90 days, with an inflection point at ≈ 74 kg (Figure 3). This pattern was consistent when body weight was analyzed categorically. Compared with the reference group (71–80 kg), patients weighing <50 kg had a significantly higher likelihood of death (1.65 [95% CI, 1.04–2.61]), but again no significant differences were observed

Table 1. Baseline Characteristics of Patients With Acute Ischemic Stroke Who Received Intravenous Alteplase by Body Weight

	≤50 kg (n=264)	51–60 kg (n=722)	61–70 kg (n=903)	71–80 kg (n=757)	>80 kg (n=560)	P value
Time from the onset of symptoms to randomization, h	2.9 [2.1–3.6]	2.8 [2.2–3.6]	2.7 [2.0–3.4]	2.7 [2.0–3.5]	2.4 [1.9–3.3]	<0.001
Demography						
Sex (male)	55/264 (20.8)	326/722 (45.2)	591/903 (65.4)	567/757 (74.9)	451/560 (80.5)	<0.001
Age, y	71.4 (12.6)	68.1 (12.4)	66.5 (12.4)	65.4 (13.1)	64.1 (12.9)	<0.001
≥80 y old	62/264 (23.5)	113/722 (15.7)	111/903 (12.3)	111/757 (14.7)	63/560 (11.3)	<0.001
Asian ethnicity	209/264 (79.2)	574/722 (79.5)	636/902 (70.5)	453/757 (59.8)	172/560 (30.7)	<0.001
Clinical features						
Systolic BP value, mm Hg	145.0 [131.0–164.0]	149.0 [130.0–163.5]	152.5 [136.5–165.0]	153.5 [137.5–167.0]	151.5 [138.0–165.0]	<0.001
Diastolic BP, mm Hg	80.0 [73.0–88.0]	82.5 [74.5–91.0]	85.0 [77.0–93.5]	87.0 [78.5–95.0]	84.8 [75.5–94.0]	<0.001
Heart rate, bpm	80.0 [70.0–88.0]	78.0 [69.0–86.0]	78.0 [69.0–86.0]	78.0 [68.0–88.0]	78.0 [68.5–88.0]	0.80
NIHSS score*	11.0 [6.0–16.0]	9.0 [5.0–14.0]	8.0 [5.0–14.0]	8.0 [5.0–13.0]	7.0 [5.0–12.0]	<0.001
GCS score†	14.0 [12.0–15.0]	15.0 [13.0–15.0]	15.0 [14.0–15.0]	15.0 [14.0–15.0]	15.0 [14.0–15.0]	<0.001
Medical history						
Hypertension	141/264 (53.4)	448/720 (62.2)	549/902 (60.9)	504/757 (66.6)	372/559 (66.5)	<0.001
Currently treated hypertension	88/264 (33.3)	304/720 (42.2)	390/902 (43.2)	369/757 (48.7)	303/559 (54.2)	<0.001
Previous stroke (ischemic, hemorrhagic, or uncertain)	40/264 (15.2)	121/722 (16.8)	183/903 (20.3)	144/757 (19.0)	86/560 (15.4)	0.07
Coronary artery disease	26/264 (9.8)	100/720 (13.9)	138/902 (15.3)	119/757 (15.7)	92/559 (16.5)	0.11
Other heart disease (valvular or other)	30/264 (11.4)	49/720 (6.8)	54/902 (6.0)	43/757 (5.7)	55/559 (9.8)	<0.01
Atrial fibrillation confirmed on ECG	54/264 (20.5)	111/720 (15.4)	133/902 (14.7)	88/757 (11.6)	73/559 (13.1)	0.007
Diabetes	37/264 (14.0)	116/720 (16.1)	163/902 (18.1)	161/757 (21.3)	150/559 (26.8)	<0.001
Hypercholesterolemia	33/264 (12.5)	75/720 (10.4)	130/902 (14.4)	146/757 (19.3)	152/559 (27.2)	<0.001
Current smoker	38/264 (14.4)	143/719 (19.9)	211/901 (23.4)	221/757 (29.2)	132/557 (23.7)	<0.001
Estimated premorbid function (mRS score)						<0.001
No symptoms (score 0–1)	128/264 (48.5)	444/722 (61.5)	584/903 (64.7)	486/757 (64.2)	367/560 (65.5)	
Symptoms without any disability (score 2–6)	136/264 (51.5)	278/722 (38.5)	319/903 (35.3)	271/757 (35.8)	193/560 (34.5)	
Medication at time of admission						
Warfarin anticoagulation	11/264 (4.2)	16/719 (2.2)	21/901 (2.3)	9/757 (1.2)	20/558 (3.6)	0.02
Aspirin or other antiplatelet agent	47/264 (17.8)	145/719 (20.2)	199/901 (22.1)	177/757 (23.4)	162/558 (29.0)	<0.001
Statin or other lipid-lowering agent	29/264 (11.0)	97/719 (13.5)	165/901 (18.3)	139/756 (18.4)	163/558 (29.2)	<0.001
Presumed stroke cause‡						
Large artery disease due to significant atheroma	99/261 (37.9)	290/715 (40.6)	376/893 (42.1)	303/746 (40.6)	184/558 (33.0)	0.009
Small vessel disease	48/261 (18.4)	148/715 (20.7)	208/893 (23.3)	159/746 (21.3)	101/558 (18.1)	0.15
Cardioembolic	76/261 (29.1)	162/715 (22.7)	160/893 (17.9)	121/746 (16.2)	108/558 (19.4)	<0.001
Weight, kg	47.0 (3.5)	57.2 (2.8)	66.8 (2.8)	76.5 (2.9)	92.0 (10.9)	<0.001

Data are n/n (%), mean (SD), or median [IQR]. P values are based on χ^2 , t test, or K-W test. BP indicates blood pressure; GCS, Glasgow Coma Scale; IQR, interquartile range; mRS, modified Rankin Scale; and NIHSS, National Institutes of Health Stroke Scale.

*NIHSS scores range from 0 to 42, with higher scores indicating more severe neurological deficit.

†Scores on the GCS range from 15 (normal) to 3 (deep coma).

‡Diagnosis according to the clinician's interpretation of clinical features and results of investigations at the time of admission to the hospital.

across the other weight categories (1.16 [95% CI, 0.80–1.68] for 51–60 kg, 0.91 [95% CI, 0.64–1.29] for 61–70 kg, and 1.07 [95% CI, 0.72–1.56] for >80 kg; Table 2). The inclusion of weight resulted in an increase in both Akaike Information Criterion and C statistic (Table S2).

There was no significant association between body weight and the likelihood of sICH, whether body weight was analyzed as a continuous or categorical variable (Figure 3; Table 2). When the analysis was performed separately by sex group, no significant interaction was observed. (Figures S2 and S3).

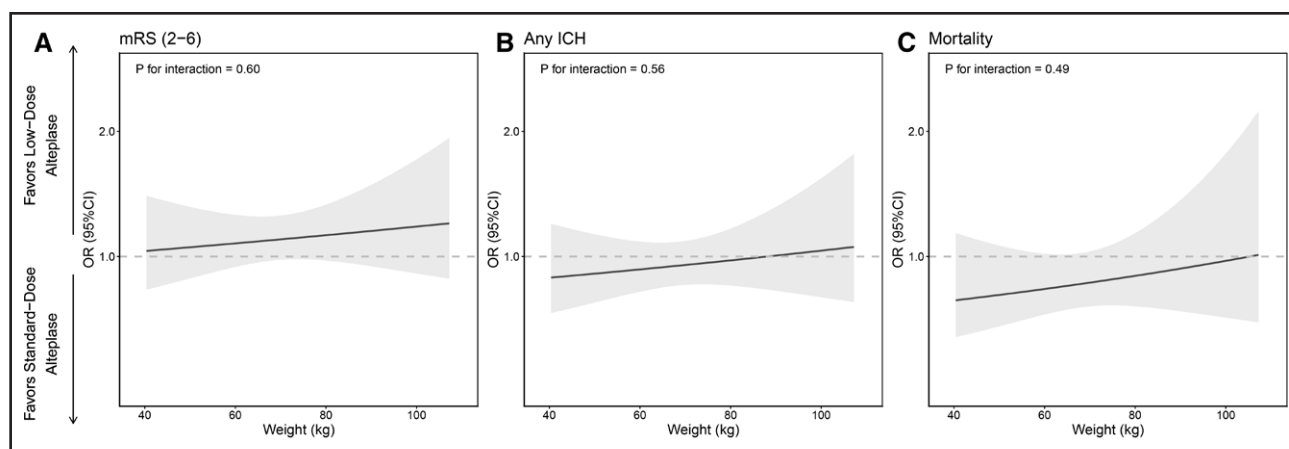


Figure 1. Randomized treatment effect on the likelihood of study outcome at day 90, by weight in logistic regression models.

A, Binary modified Rankin Scale (mRS) score outcome, adjusted for age and baseline National Institutes of Health Stroke Scale (NIHSS) score. **B**, Any intracerebral hemorrhage (ICH) outcome, adjusted for age and baseline NIHSS score. **C**, Mortality outcome, adjusted for age and baseline NIHSS score. The solid line represents the estimated odds ratio (OR) of outcome across the range of body weight, and the shaded area indicates 95% CIs. An OR of 1.0 indicates no difference between treatment groups; values >1.0 indicate higher odds of poor outcome with low-dose alteplase, whereas values <1.0 indicate lower odds.

DISCUSSION

In this post hoc analysis of ENCHANTED, we found that the treatment effects of low-dose versus standard-dose alteplase on clinical outcomes were similar irrespective of the body weight of patients. However, patients with a low body weight had a worse functional outcome and a greater likelihood of death without any increase in sICH after receiving thrombolysis treatment for AIS.

Our findings provide further insights into the role of body weight in relation to the effects of thrombolysis for AIS. Although previous observational studies have raised concerns that obesity may attenuate the effectiveness of alteplase,²⁷ potentially due to under-dosing related to the 90 mg dose cap, other studies have found no significant association between body weight and treatment outcomes.¹² Evidence in low body weight patients remains sparse due to their underrepresentation in existing studies.¹³ However, these observational analyses are limited by selection bias and measured and unmeasured confounding. Being based on a large, prospectively and systematically collected database, our analysis of the ENCHANTED randomized controlled trial overcomes many of these limitations. We found no evidence that body weight modifies the comparative efficacy or safety of low-dose versus standard-dose alteplase. This result supports the use of current weight-based dosing protocols without the need for adjustment across a broad weight spectrum. Our findings fill a key evidence gap and offer reassurance to clinicians who want to optimize thrombolysis strategies in a broad range of patient characteristics.

Low body weight patients in our cohort experienced the worst outcomes after thrombolysis, including higher

mortality and worse functional recovery. Low body weight appears to be an independent predictor of poor prognosis in AIS,²⁸ likely reflecting underlying frailty, malnutrition, and limited physiological reserve. These patients are more vulnerable to the hypercatabolic stress of stroke due to insufficient energy stores to support their recovery,^{29,30} along with the common presence of comorbid conditions, such as malignancy and chronic disease. Low body weight may, therefore, act as a surrogate for frailty or chronic illness, contributing to greater disability and mortality.^{31,32} One large registry found that underweight stroke patients had ~50% higher odds of significant disability and nearly double the short-term mortality risk compared with those of normal weight.²⁹ These findings reinforce the prognostic importance of body habitus in AIS and reflect the obesity paradox, where being underweight rather than excessively overweight is associated with a significantly worse outcome.^{28,31}

Low body weight patients in our cohort did not show any increase in sICH after intravenous alteplase, consistent with larger registry data reporting similar hemorrhage rates across weight groups.¹³ This suggests that standard weight-based alteplase dosing (0.9 mg/kg) provides appropriate fibrinolytic activity even in low body weight individuals, and avoids the possibility of inadvertent overdosing.¹³ Although frailty and malnutrition associated with low weight could theoretically increase sICH risk,²⁸ weight itself has not been identified as an independent predictor of sICH after adjusting for confounders.¹³ Instead, factors such as age, baseline BP, and stroke severity are more relevant determinants of intracerebral hemorrhage.³³ These findings are reassuring; low body weight alone should not be a contraindication to thrombolysis, as low body weight patients demonstrate comparable safety outcomes to those who are of normal weight.

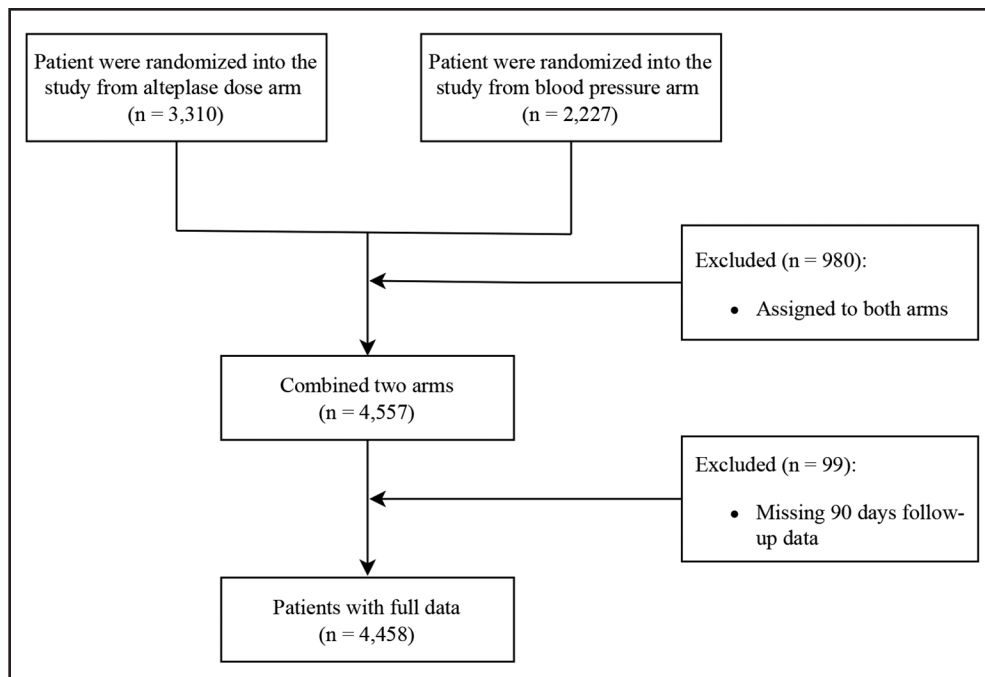


Figure 2. Flowchart of participants from the combined alteplase-dose arm and blood pressure arm.

Key strengths of this study are the pragmatic design and large sample of patients with a broad range of characteristics who were recruited from a variety of healthcare settings worldwide, which enhance the generalizability of the results. The high rates of adherence to treatment and follow-up allowed rigorous reporting of outcomes and serious adverse events, whereas the central adjudication of sICH to standard definitions allowed a reliable assessment of the balance of benefits and harms of alteplase treatment. However, our study has some limitations: most of the participants had only mild degrees of neurological severity and received study treatment with a slightly

longer delay from symptom onset than has occurred in other trials and quality assurance studies of alteplase in AIS.¹ ENCHANTED was undertaken before endovascular thrombectomy became standard practice for the treatment of AIS due to large-vessel occlusion. We did not collect data on the height of patients, so we could not undertake analysis by body mass index or other anthropometric indices. Moreover, body weight was obtained according to routine clinical practice and not standardized, introducing potential misclassification that may have affected both alteplase dosing and outcome associations. As this was a retrospective, nonprespecified

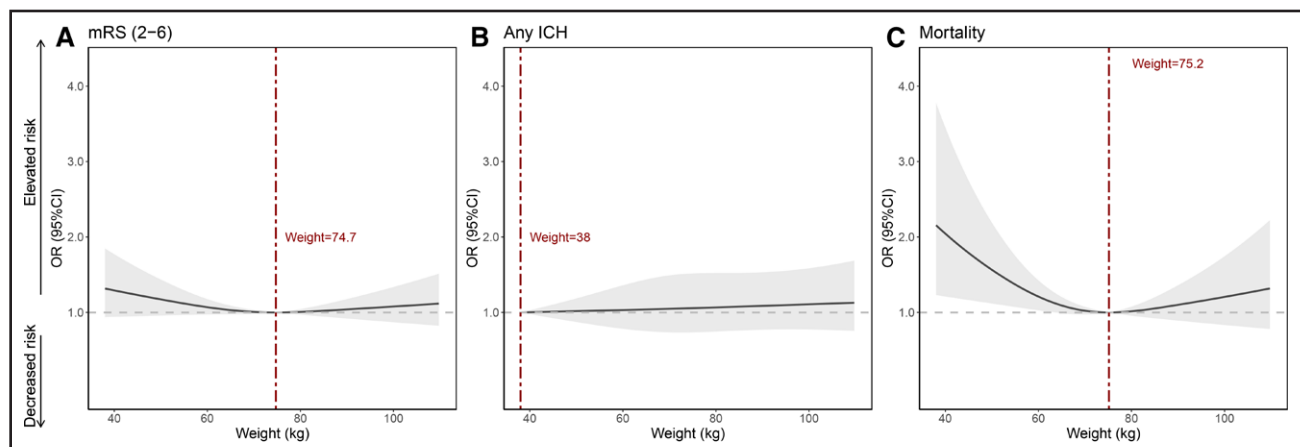


Figure 3. Restricted cubic spline plot of adjusted odds ratio (OR, of outcome; 95% CI) for the association between weight and study outcomes on day 90.

Models were adjusted for randomized treatment allocation; time from onset to randomization; age; sex; ethnicity; baseline systolic blood pressure; baseline National Institutes of Health Stroke Scale score; history of stroke, hypertension, atrial fibrillation, diabetes, valvular heart disease, and hypercholesterolemia; current treated hypertension; current smoking status; premorbid use of anticoagulants, other antiplatelet agents, and statins or other lipid-lowering therapies; presumed stroke cause. ICH indicates intracerebral hemorrhage; and mRS, modified Rankin Scale.

Table 2. Categorized Baseline Body Weight and 90-Day Outcome After Thrombolysis of Acute Ischemic Stroke

Outcome	Weight category, kg	No. of patients, n	Events, n (%)	OR (95% CI)	aOR (95% CI)	Adjusted <i>P</i> _{interaction}
mRS score (2–6)	≤50	394	238 (60.4%)	1.64 (1.30–2.08)	1.41 (1.06–1.87)	0.22
	51–60	998	494 (49.5%)	1.05 (0.88–1.25)	1.04 (0.84–1.27)	
	61–70	1257	627 (49.9%)	1.07 (0.91–1.26)	1.13 (0.94–1.36)	
	71–80	1035	499 (48.2%)	1.00 (Reference)	1.00 (Reference)	
	>80	774	373 (48.2%)	1.00 (0.83–1.20)	1.06 (0.86–1.32)	
Mortality	≤50	394	52 (13.2%)	1.49 (1.03–2.14)	1.65 (1.04–2.61)	0.69
	51–60	998	99 (9.9%)	1.10 (0.82–1.49)	1.16 (0.80–1.68)	
	61–70	1257	105 (8.4%)	0.93 (0.70–1.25)	0.91 (0.64–1.29)	
	71–80	1035	92 (8.9%)	1.00 (Reference)	1.00 (Reference)	
	>80	774	68 (8.8%)	0.99 (0.71–1.37)	1.07 (0.72–1.56)	
Any ICH	≤50	394	80 (20.3%)	1.21 (0.90–1.62)	1.00 (0.71–1.40)	0.90
	51–60	998	193 (19.3%)	1.14 (0.91–1.43)	1.07 (0.83–1.38)	
	61–70	1257	235 (18.7%)	1.09 (0.88–1.35)	1.07 (0.85–1.35)	
	71–80	1035	180 (17.4%)	1.00 (Reference)	1.00 (Reference)	
	>80	774	140 (18.1%)	1.05 (0.82–1.34)	1.19 (0.91–1.55)	
ICH on NINDS criteria	≤50	394	30 (7.6%)	1.17 (0.74–1.81)	0.97 (0.57–1.59)	0.60
	51–60	998	65 (6.5%)	0.99 (0.70–1.41)	0.88 (0.59–1.31)	
	61–70	1257	95 (7.6%)	1.16 (0.84–1.61)	1.08 (0.76–1.53)	
	71–80	1035	68 (6.6%)	1.00 (Reference)	1.00 (Reference)	
	>80	774	47 (6.1%)	0.92 (0.62–1.35)	1.07 (0.71–1.60)	
ICH on ECASS2 criterion	≤50	394	23 (5.8%)	1.54 (0.90–2.59)	1.43 (0.77–2.61)	0.24
	51–60	998	41 (4.1%)	1.07 (0.68–1.67)	0.98 (0.59–1.62)	
	61–70	1257	64 (5.1%)	1.33 (0.90–2.01)	1.24 (0.80–1.94)	
	71–80	1035	40 (3.9%)	1.00 (Reference)	1.00 (Reference)	
	>80	774	28 (3.6%)	0.93 (0.57–1.52)	1.07 (0.63–1.80)	

Models were adjusted for randomized treatment allocation; time from onset to randomization; age; sex; ethnicity; baseline SBP; baseline NIHSS score; history of stroke, hypertension, atrial fibrillation, diabetes, valvular heart disease, and hypercholesterolemia; current treated hypertension; current smoking status; premorbid use of anticoagulants, other antiplatelet agents, and statins or other lipid-lowering therapies; presumed stroke cause. aOR indicates adjusted odds ratio; ECASS2, European Cooperative Acute Stroke Study; ICH, intracranial hemorrhage; mRS, modified Rankin scale; NIHSS, National Institutes of Health Stroke Scale; NINDS, National Institute of Neurological Disorders and Stroke; OR, odds ratio; and SBP, systolic blood pressure.

analysis, the findings are susceptible to chance associations, residual confounding, and selective reporting bias. Furthermore, as a post hoc exploratory analysis, the study is likely underpowered to detect treatment-covariate interactions, for which substantially larger sample sizes are typically required. In addition, multiple statistical comparisons, including subgroup and interaction analyses, were performed without formal adjustment, increasing the risk of type I error; therefore, the findings should be interpreted with caution.

In summary, this post hoc analysis of the ENCHANTED trial confirms that body weight did not alter the comparative effects of low-dose compared with standard-dose alteplase in patients with AIS. Low body weight was associated with a poor functional outcome and mortality but did not increase the likelihood of sICH. The absence of excess intracerebral hemorrhage risk in patients with low body weight is not only reassuring to stakeholders but indicates that low body weight alone should not deter the use of intravenous thrombolysis; Lower body weight was associated with comparable safety outcomes

to being of normal weight. These analyses support the appropriateness of current weight-based alteplase dosing in AIS.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S2
Figures S1–S3
STROBE Checklist

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